ÇANKAYA UNIVERSITY
FACULTY OF ENGINEERING

COMPUTER ENGINEERING DEPARTMENT

Implementation Document

CENG 408
Innovative System Design and Development II

A SMART HOME SYSTEM

Enis Mirsad Şengül
202211065
Duhan Ayberk Seven
202211062
Deniz Arda Çınarer
202211019
İbrahim Ersan Özdemir
202211054

Advisor: Prof. Dr. Murat Koyuncu

Component	Technology / Implementation
Edge Device	Raspberry Pi 5 (8 GB RAM)
Backend	FastAPI + Supabase
Frontend	React 19 + Vite
Mobile Application	Flutter
AI Pipeline	MediaPipe + dlib Face Recognition
Streaming	WebSocket Relay Architecture
Database	Supabase PostgreSQL

Conclusion

The Smart Home System project successfully achieved its primary objective of developing an integrated IoT platform combining environmental monitoring, AI-powered security, realtime communication, and multi-platform accessibility within a unified architecture.

During the development lifecycle, the project evolved from a prototype into a production-oriented distributed system consisting of a Raspberry Pi 5 edge controller, a FastAPI cloud gateway, a Supabase-based backend infrastructure, a React web dashboard, and a Flutter mobile application.

One of the strongest aspects of the system was the implementation of an edge-first architecture. Instead of transferring all workloads to the cloud, the Raspberry Pi performed local sensor processing and face recognition directly on the device. This approach reduced latency, minimized bandwidth consumption, and improved overall system resilience.

The AI-based face recognition pipeline became one of the major technical contributions of the project. By combining MediaPipe face detection with a fallback dlib-based recognition mechanism, the system achieved stable recognition performance on ARM-based hardware under standard indoor conditions.

> “More importantly, the project evolved beyond a simple academic prototype and became a production-oriented distributed IoT platform capable of solving real-world smart home monitoring and security problems.”

Key Achievements

- Realtime sensor monitoring using Supabase Realtime subscriptions
- Hybrid AI face recognition pipeline with MediaPipe and dlib
- On-demand WebSocket live streaming architecture
- Offline-resilient synchronization system using SQLite queueing
- Role-based authentication and cloud synchronization
- Cross-platform support through React web and Flutter mobile applications
- Stable long-duration runtime with realtime synchronization

System Stability and Testing

System testing demonstrated that the platform can operate continuously for extended durations under real-world conditions. During development and evaluation, the system remained active for sessions lasting up to approximately 12 hours continuously and was actively tested throughout nearly two months of iterative development. Although a small number of crashes were observed during early iterations, these issues were significantly reduced through debugging, synchronization improvements, and architectural refinements. By the final implementation stage, the realtime infrastructure and core monitoring systems were operating in a stable and reliable manner.

Future Work

Despite the successful implementation of the current system, several opportunities remain for future improvement and expansion.

Future Area	Description
Multi-Camera Support	Support for multiple synchronized camera streams and centralized monitoring.
Smart Automation	Implementation of advanced automation scenarios such as Away Mode and Night
Energy Monitoring	Integration of power consumption sensors for realtime energy analytics.
Voice Assistant Integration	Compatibility with Google Assistant, Alexa, or local voice assistants.
Additional Sensors	Support for CO, air quality, vibration, and water leakage sensors
Security Improvements	Migration from ws:// to wss:// with TLS encryption and stronger device authentication
AI Improvements	Use of Edge TPU acceleration and ANN indexing methods such as FAISS

In conclusion, this project demonstrates the feasibility of building a modern, intelligent, and distributed smart home ecosystem using affordable hardware and contemporary cloud technologies. The platform combines IoT sensor integration, edge-based artificial intelligence, realtime communication, and cross-platform accessibility into a cohesive and functional system. The experience gained throughout the development process provided valuable insight into distributed systems, embedded software engineering, computer vision, cloud infrastructure, and realtime application development, establishing a strong foundation for future academic and professional work.